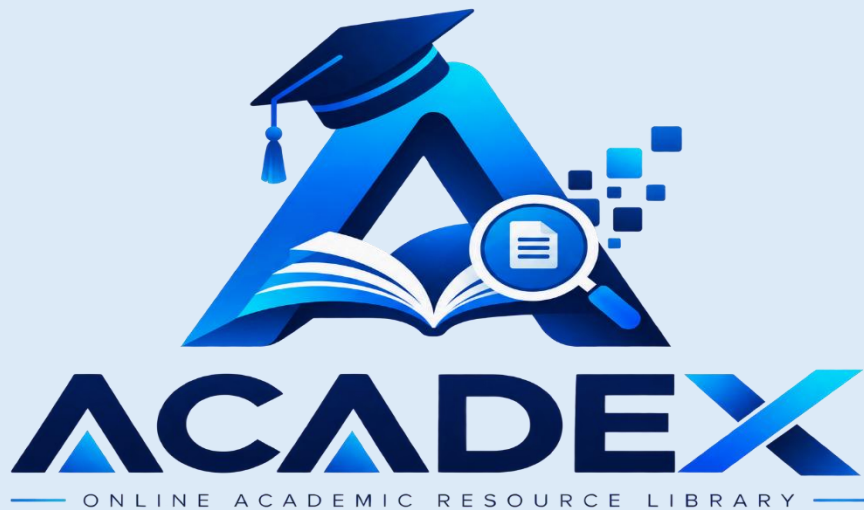




ACADEX

ONLINE ACADEMIC RESOURCE LIBRARY

PROJECT PROPOSAL

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1. Project Purpose

Acadex is an online academic resource library proposed by that will serve students as an integrated resource for discovering, accessing, organizing and sharing academic resources. The system will provide a digital environment that will enable students to access academic content without being solely reliant on a physical library or using multiple platforms unrelated to the academic content.

Acadex will employ these principles in a student-centered academic resource setting. The project aims to develop a well-organized and user-friendly resource platform for students to easily access and engage with academic materials, inspired by academic resource sharing websites like Studocu. The primary goal of Acadex is thus to make academic resources more accessible, organized and discoverable via a single online platform. The system will aim to:

- Provide students with a central location for academic resources.
- Allow students to search for resources using keywords, subjects and courses.
- Allow registered users to upload and share academic resources.
- Organize resources into appropriate subjects, courses and categories.
- Allow users to view, download and save useful resources.
- Provide ratings and comments to help users evaluate resources.
- Allow users to report inappropriate, inaccurate or unsuitable content.
- Provide administrators with tools to manage users and resources.
- Create a responsive interface that can be accessed using computers, tablets and mobile devices.
- Provide a system architecture that can be expanded with additional functionality in the future.

2. Proposed system and features

Acadex will function as an online academic resource repository with additional community and resource-management features. The system will focus on making academic resources easier to discover, organize and interact with.

2.1 User registration and authentication

The users will be able to register and log in to Acadex in a secure way. Registered users will have access to features such as uploading resources, bookmarking resources, commenting and managing their profiles.

User authentication will also enable the system to differentiate between normal users and administrators. This will give administrators more user, resource and reported content management privileges.

2.2 Academic resource repository

Academic resources like lecture notes, study guide, summaries, old test papers etc. will be stored in the system. Each resource will contain information such as its title, description, subject, course, resource type and uploader.

The concept of online library environment of different information resources is adopted for this approach. Mr Chalukya in the journal provided as a source highlights the significance of the online library services in providing easy access to information resources and services to the users (Chalukya, 2021).

2.3 Search and filtering

One of the most crucial features of Acadex will be search. Users will be able to search for resources by keywords and restrict their search by filtering the results by the following, *Subject, Course, Resource Type, Institution, Date, Uploaded* and *Rating*.

Existing library discovery systems have an impact on the proposed search functionality. For instance, WorldCat gathers all library collections into a single search environment and offers users search and filtering tools to help focus the results. WorldCat also offers advanced search options and facets for narrowing down searches, which will be beneficial for designing the search experience for Acadex (Anon., 2026).

2.4 Ratings and Comments

Users will have the ability to rate and leave comments for resources. This will enable students to give feedback on the effectiveness and quality of resources.

These ratings and comments will also enable users to identify resources that other students have found useful. This gives a sense of community interaction in Acadex, rather than using it as a document storage system.

Some well-known educational resource sites, like OER Commons, also offer features like ratings, comments, and resource evaluation, making it an opportunity for users to engage with educational resources (Anon., 2026).

2.5 Content reporting and moderation

Users will have the ability to report inappropriate, misleading, duplicated or otherwise unsuitable resources. Administrators will then have the ability to see the content that has been reported and make a decision on what action to take.

Content is a significant issue for Acadex since they will be permitting the addition of resources by the user. All documents uploaded to an electronic file do not automatically imply that they are accurate, appropriate or suitable for educational purposes.

Educational resource platforms like OER Commons offer mechanisms for submitting, evaluating and reporting resources, highlighting the need for a mechanism to control the process of managing user submitted educational content (Anon., 2026).

2.6 Administration

An administrator dashboard will be created to control users, resources, categories and reports. The administrators will be able to:

- Manage registered users.
- Review uploaded resources.
- Remove inappropriate content.
- Review reported resources.
- Organize topics and types.
- Monitor system activity.
- Keep the quality of the platform at an adequate level.

This will assist in maintaining the organization and appropriateness of Acadex for academic purposes.

3. Implementation technologies

Component	Technology	Purpose
Frontend	HTML5, CSS3, JavaScript	User interface and interaction
UI Framework	Bootstrap	Responsive and consistent design
Backend	PHP	Application logic and server side processing
Database	MySQL	Storage and management of system data
Development Environment	XAMPP	Local development and testing
Web Server	Apache	Running the web application

4. Software Development Life Cycle Model

Acadex is going to be developed using the Agile Software Development Life Cycle (SDLC) model. Agile is suitable because the project has a number of features which can be developed and tested in incremental steps. The project group will not strive for a full development at once, but will break it up into smaller chunks (Mishra, 2023).

The first prototype can be limited to the principle functionality of the platform, like the creation of databases, the enrollment and authentication of users. Later iterations may add uploading of resources, searching, categorization, downloading, ratings and comments, then you can take advantage of advanced capabilities like reporting, and administration (Liyanage, 2025).

The proposed development process will consist of:

- **Planning:** The project group will identify the target users, problems, objectives and functional requirements of Acadex.
- **System design:** The database, user interface, navigation, system architecture and overall functionality will be designed before implementation.
- **Development:** The system will be developed using PHP, MySQL, HTML5, CSS3, JavaScript and Bootstrap.
- **Testing:** Individual features and the complete integrated system will be tested to identify errors, security issues and usability problems.
- **Evaluation and Feedback:** The system will be evaluated, and feedback will be used to identify areas that require improvement.
- **Deployment:** The completed system will be prepared for its intended environment and made available to its users.
- **Maintenance and Enhancement:** Errors will be corrected and additional features may be introduced where necessary.

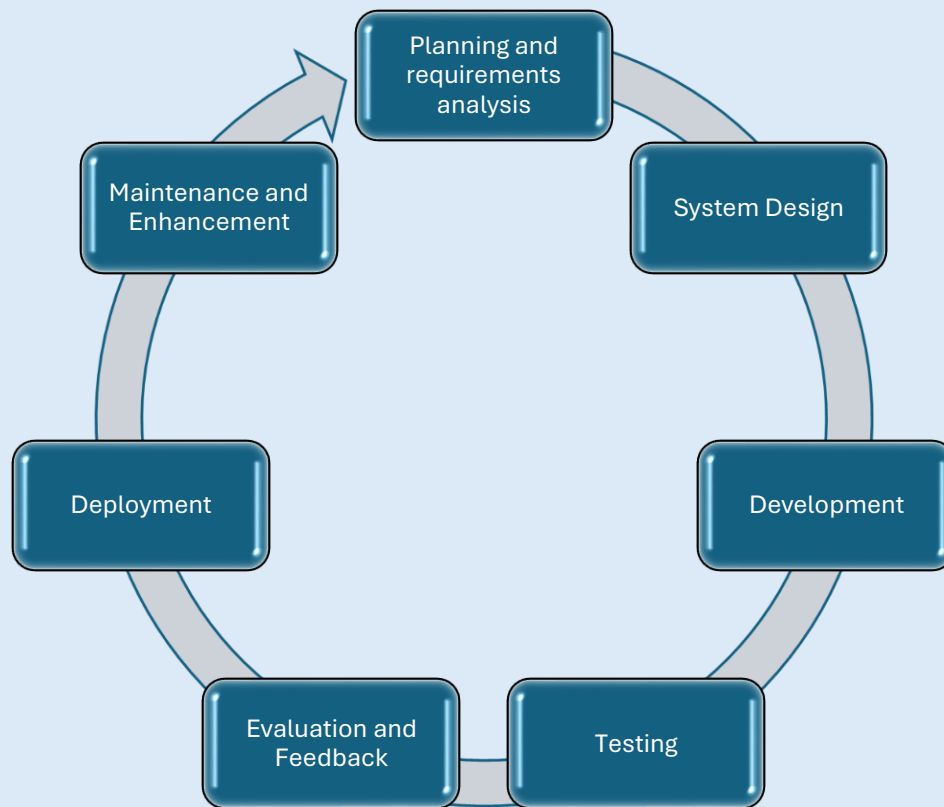


Figure 1 Rotation of phases in the development (Mishra, 2023)

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